OntoCoDec: Type-Valid Information Extraction via Compiled Ontology Constraints

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Abstract

Large language models applied to information extraction (IE) produce systematic ontology constraint violations—generating triples with entity types incompatible with declared domain/range specifications. We present OntoCoDec, a framework that automatically compiles IE ontology schemas into composable decoding constraints enforced at generation time. OntoCoDec operates in two layers: a *grammar layer* (context-free grammar enforcing JSON structure and type vocabulary) and a *semantic layer* (stateful entity registry with dynamic logit masking enforcing domain/range constraints). The two layers compose via bitwise AND of token masks with an enabling dependency: grammar ensures structural validity, enabling semantic constraints to enforce type rules. On SciERC, OntoCoDec reduces domain/range violations from 15.4% to 0.93% (↓94%) with F1-neutral impact (+0.13pp, p=0.58). On DocRED, violations decrease by 44.7% (Qwen) and 90.2% (Llama-3.1-8B) with statistically significant F1 improvements (+0.50pp, p¡0.001). Cross-model validation on three architectures confirms zero-code generalization. We identify cross-field consistency violations (CF=9.9% on SciERC) as an inherent limitation of per-relation independent enforcement and provide root-cause analysis tracing CF to type-system boundary ambiguity.

1 Introduction

LLM-based information extraction systems promise flexible, schema-driven extraction without task-specific training data. However, when prompted to extract structured knowledge according to an ontology schema, LLMs systematically violate type constraints—generating triples where entity types are incompatible with the declared domain and range of relation types. These violations are not random: on SciERC (7 relation types), 15.4% of generated triples violate domain/range constraints, with violation rates ranging from 0.8% (Used-for) to 42.4% (Compare), driven by constraint tightness rather than ontology size.

Prompt-level intervention (explicitly describing type constraints in the system prompt) reduces violations by 62.4% on SciERC but only 31.5% on DocRED (96 relations), demonstrating that prompt-based approaches scale poorly with ontology complexity. Worse, prompt intervention is unreliable at the per-relation level: for Evaluate-for, schema-informed prompting *increases* the violation rate from 24.1% to 29.9% due to prior-constraint conflict.

We propose OntoCoDec, a framework that compiles ontology schemas into two complementary decoding constraint layers:

1. A **grammar layer** enforcing JSON structure and restricting generated type tokens to the valid vocabulary via context-free grammar (implemented with XGrammar).

2. A **semantic layer** maintaining a stateful entity registry during generation and dynamically masking logits to enforce domain/range compatibility.

The layers compose via bitwise AND with an *enabling dependency*: grammar ensures every relation type token is in the valid vocabulary, which enables the semantic layer to look up and enforce domain/range rules. Without grammar, the semantic layer encounters unknown types and cannot enforce constraints (semantic-only achieves only $\downarrow$24% vs full composition $\downarrow$94%).

We position OntoCoDec as the first decidability-aware type enforcement system for autoregressive IE. Unlike prior grammar-constrained decoding work (SynCode, XGrammar) that handles context-free structural constraints, OntoCoDec addresses prefix-decidable stateful constraints—a fundamentally different decidability class requiring runtime state (entity registry) beyond what CFGs can express.

Our contributions:

- A systematic violation landscape study quantifying ontology constraint violations across models, datasets, and prompt strategies.
- A composable dual-layer constraint framework with automatic schema compilation, requiring zero code changes across models.
- Empirical validation on 2 datasets $\times$ 3 models showing DR$\downarrow$44–99% at F1-neutral or positive impact.
- Analysis of the DR–CF trade-off as an inherent property of per-relation enforcement, with root-cause attribution.

2 Background

2.1 Grammar-Constrained Decoding

Grammar-constrained decoding restricts autoregressive generation to outputs conforming to a formal grammar. Given vocabulary V and grammar G, at each step t the decoder produces logits over V; a constraint mask $M_t \subseteq V$ zeros out tokens that would lead to strings outside $L(G)$. XGrammar compiles context-free grammars into efficient token masks via adaptive tokenization. This guarantees structural validity (valid JSON, correct field names) but cannot express semantic constraints requiring runtime state.

2.2 IE Ontology Constraints

An IE ontology $\mathcal{O} = (E, R, \text{dom}, \text{rng})$ defines entity types E, relation types R, and functions dom : $R \to 2^E$, rng : $R \to 2^E$ specifying valid argument types. A triple (h, r, t) with type$(h) \notin$ dom(r) or type$(t) \notin$ rng(r) constitutes a *domain/range violation* (DR). We additionally define relation vocabulary violations (RV: $r \notin R$), entity type violations (ET: type$(e) \notin E$), and cross-field consistency violations (CF: same entity assigned different types across triples).

2.3 Constraint Decidability Taxonomy

We classify IE constraints by prefix-decidability:

(a) **Purely local**: Entity type $\in$ vocabulary (grammar layer; CFG-expressible).

(b) **Prefix-decidable with state**: Domain/range compatibility requires knowing previously generated entity types (semantic layer; requires entity registry).

(c) **Truly non-local**: Forward coreference consistency—out of scope for forward-only decoding.

This taxonomy distinguishes OntoCoDec from prior work operating only at level (a).

3 Method

3.1 Schema Compilation

Given ontology $\mathcal{O}$, OntoCoDec's compiler produces:

1. A **context-free grammar** $G_{\mathcal{O}}$ encoding valid JSON structure with entity type tokens restricted to E and relation type tokens restricted to R.

2. A **constraint table** $T_{\mathcal{O}}$ mapping each $r \in R$ to its valid domain/range type sets.

3.2 Grammar Layer

The grammar layer applies $G_{\mathcal{O}}$ via XGrammar, producing mask M_t^{gram} at each step. This eliminates RV (relation vocabulary violations) and ET (entity type violations) by construction.

3.3 Semantic Layer: TypeConstraintProcessor

The semantic layer maintains an *entity registry* $\mathcal{R}$ mapping entity mentions to their assigned types during generation. When the decoder reaches a position requiring an entity type token within a relation triple for relation r:

- If generating the head entity type: $M_t^{\text{sem}} = \{e \in E \mid e \in \text{dom}(r)\}$
- If generating the tail entity type: $M_t^{\text{sem}} = \{e \in E \mid e \in \text{rng}(r)\}$

The registry enables cross-reference: if an entity was previously typed, the semantic layer constrains subsequent mentions to compatible types.

3.4 Composition and Graceful Degradation

The final mask is $M_t = M_t^{\text{gram}} \cap M_t^{\text{sem}}$ (bitwise AND). When $M_t^{\text{gram}} \cap M_t^{\text{sem}} = \emptyset$ (constraint intersection empty due to high tightness), the system falls back to grammar-only: $M_t = M_t^{\text{gram}}$. This guarantees structural validity as a floor.

Fallback frequency quantifies *constraint coverage*: on SciERC, $\sim$10.7 fallbacks/document; on DocRED, $\sim$57 fallbacks/document (5.3$\times$ higher), directly explaining the residual DR gap between datasets.

3.5 Decoding Strategy S1 (Single-Pass)

In strategy S1, both layers are active throughout generation. Entities are registered as they are generated; subsequent relation triples benefit from the accumulated registry. This is our primary strategy.

4 Experiments

4.1 Violation Landscape (Prompt Comparison)

We measure baseline violation rates with minimal and schema-informed prompts on SciERC and DocRED using Qwen3-8B.

Per-relation analysis reveals violation concentration: Compare (42.4%), Feature-of (33.5%), and Part-of (25.0%) account for most SciERC violations, while Used-for (0.8%) and Conjunction (0%) are nearly

Dataset	Prompt	DR%	RV%	ET%
SciERC	Minimal	15.4	1.4	0.4
	Schema-informed	5.8	0.3	0.0
DocRED	Minimal	18.7	28.9	7.5
	Schema-informed	12.8	18.4	0.6

Table 1: Violation rates across prompt strategies. Schema-informed prompting reduces DR by 62.4% on SciERC but only 31.5% on DocRED, demonstrating poor scaling with ontology complexity.

Configuration	DR%	RV%	CF%	R-F1%
Unconstrained	15.4	1.4	0.1	8.23
Grammar-only	14.8	0.0	0.1	8.23
Semantic-only	11.7	1.5	2.7	8.15
Full S1	**0.93**	**0.0**	**9.9**	**8.36**

Table 2: SciERC layer ablation. Full S1 achieves DR↓94% with F1-neutral impact (+0.13pp, p=0.58, 95% CI [−0.35, +0.63]). Grammar-only eliminates RV/ET but not DR; semantic-only achieves only ↓24% DR due to OOV relation types preventing constraint lookup.

violation-free. This distribution correlates with constraint tightness—relations permitting fewer type combinations exhibit higher violation rates.

Schema-informed prompting on SciERC's Evaluate-for *increases* violations from 24.1% to 29.9% (+5.8pp), demonstrating prior-constraint conflict: the model's distributional prior for plausible type assignments actively interferes with prompt-level constraint signals.

4.2 SciERC Ablation Study

The ablation confirms the enabling dependency: grammar-only reduces DR by only 0.6pp (structural constraints alone cannot enforce type semantics), while semantic-only achieves ↓24% (hampered by OOV types). Full composition achieves ↓94% because grammar eliminates OOV, enabling semantic enforcement on all triples.

CF rises from 0.1% (unconstrained) to 9.9% (full S1). Root-cause analysis identifies the dominant pattern: Generic↔OtherScientificTerm (60 occurrences), Material↔OtherScientificTerm (28), Material↔Method (20)—reflecting genuine type-system boundary ambiguity in SciERC rather than method failure. CF is 100% attributable to the semantic layer (grammar-only CF=0.1%).

4.3 DocRED Ablation Study

Grammar-only produces a paradoxical DR increase on DocRED: eliminating 5,189 OOV relation predictions (RV 29.7%→0.03%) forces these triples into valid relation names whose domain/range constraints they violate, making latent violations visible. From this grammar-enforced baseline (26.2%), the semantic layer achieves ↓61% additional reduction.

DocRED R-F1 improves significantly (+0.50pp, p¡0.001, 95% CI [+0.23, +0.81]) and E-F1 improves +1.23pp (p¡0.001, 95% CI [+0.94, +1.55]). Mechanism: grammar eliminates OOV relation names and non-standard entity types; under all-set evaluation, OOV predictions count as false positives, so their elimination directly increases precision.

Config	DR%	RV%	CF%	R-F1%	E-F1%
Unconstr.	18.5	29.7	0.1	7.74	58.15
Gram-only	26.2	0.03	0.1	8.31	59.38
Sem-only	13.8	29.9	1.1	7.70	57.81
Full S1	**10.2**	**0.03**	**2.6**	**8.23**	**59.37**

Table 3: DocRED (Qwen) ablation. Grammar-only DR *increases* (26.2% vs 18.5%) due to denominator change: eliminating 5,189 OOV triples exposes latent DR violations in newly valid triples. Full S1 achieves DR↓44.7% with R-F1 +0.50pp (p¡0.001).

Model	Dataset	$\text{DR\%}_u$	$\text{DR\%}_c$	↓%	ΔR-F1
Qwen-8B	SciERC	15.4	0.93	94.0	+0.13
Qwen-8B	DocRED	18.5	10.2	44.7	$+0.50^{***}$
Llama-8B	SciERC	13.3	0.08	99.4	−0.14
Llama-8B	DocRED	26.1	2.55	90.2	$+0.29^{*}$
Mistral-7B	SciERC	56.9	13.4	76.5	+0.00
Mistral-7B	DocRED	57.4	33.2	42.2	+2.02

Table 4: Cross-model DR reduction. OntoCoDec generalizes across model families with zero code modification. Subscripts: u=unconstrained, c=constrained. $^{***}p$¡0.001, $^{*}p$¡0.05.

4.4 Cross-Model Validation

OntoCoDec generalizes across model families without code modification. The DR reduction gap between Qwen DocRED (↓44.7%) and Llama DocRED (↓90.2%) is explained by the tightness×RV-baseline interaction: Qwen's high RV baseline (29.7%) means grammar forces many OOV triples into valid relation names, exposing latent DR violations; Llama's lower RV (16.0%) produces less exposure, allowing higher net reduction.

Mistral-7B exhibits catastrophically high baseline violations (DR=56.9%) indicating unsuitability for zero-shot IE, yet OntoCoDec still guarantees RV/ET→0% and achieves DR↓76.5% (SciERC), demonstrating robustness even on poorly-performing models.

4.5 Post-Hoc Filtering Comparison

Post-hoc filtering achieves higher R-F1 than OntoCoDec in the zero-shot regime because removing violating triples (many of which are false positives) boosts precision. However, post-hoc filtering: (1) loses 48.1% of DocRED triples including 127 true positives (11.2% of correct extractions); (2) cannot resolve entity-type violations (residual ET=0.4% SciERC, 2.47% DocRED); (3) requires post-processing incompatible with streaming. OntoCoDec's recall is consistently higher (SciERC: 11.60 vs 10.68; DocRED: 10.02 vs 8.40), validating the "prevent vs detect" paradigm.

4.6 Fine-Tuning Ablation

Fine-tuning (LoRA, 3 epochs) reduces violations from 15% to ∼1–2%, but residuals persist (22 violations on Qwen SciERC, 12 on Llama SciERC). Constrained decoding eliminates these residuals to 0% (or near-0%) with negligible F1 impact (<0.1pp across all configurations). On Llama DocRED, constrained decoding additionally provides R-F1 +0.51pp and E-F1 +1.30pp even in the FT regime, via elimination of residual OOV predictions.

Dataset	Method	R-F1%	Recall	Reject%
SciERC	Post-hoc	8.63	10.68	16.8
	OntoCoDec	8.36	11.60	—
DocRED	Post-hoc	9.61	8.40	48.1
	OntoCoDec	8.23	10.02	—

Table 5: Post-hoc filtering achieves higher F1 via precision gains from triple removal, but loses 48.1% of triples on DocRED (127 true positives lost). OntoCoDec preserves recall by preventing violations at generation time rather than detecting them post-hoc.

Model	Condition	DR%	CF%	R-F1%
Qwen SciERC	FT unconstr.	1.95	0.0	43.86
	FT + constr.	0.0	1.2	43.77
Llama SciERC	FT unconstr.	1.11	0.0	45.50
	FT + constr.	0.09	0.5	45.50
Llama DocRED	FT unconstr.	0.49	—	43.72
	FT + constr.	0.20	—	44.23
Qwen DocRED	FT unconstr.	0.76	—	40.86
	FT + constr.	0.01	—	40.56

Table 6: Fine-tuning ablation. FT reduces violations to 1–2%; constrained decoding eliminates residuals to ≈0% with F1-neutral impact. The two approaches are complementary: FT reduces violation volume, constraints guarantee elimination.

Fine-tuning and constrained decoding are complementary: fine-tuning provides statistical violation reduction through learned type associations, while constrained decoding provides the hard guarantee of zero residual violations regardless of fine-tuning quality.

4.7 Latency and S2 Strategy

Constrained decoding overhead is $\leq$5.4% wall-clock time (measured on Llama DocRED, 23,926s total for 907 documents). The overhead is minimal because mask computation operates on pre-compiled constraint tables with O(1) lookup.

Strategy S2 (two-pass: entity extraction followed by relation generation with input-dependent grammar) achieves comparable results to S1 on SciERC (DR=0.97%, CF=8.0%, R-F1=8.03%) but does not meaningfully reduce CF (-1.87pp, not significant) while adding inference complexity. We recommend S1 as the primary strategy.

4.8 Statistical Methodology

All F1 comparisons use bootstrap confidence intervals (10,000 iterations). SciERC R-F1: +0.13pp, p=0.58, 95% CI [-0.35, +0.63]. DocRED R-F1: +0.50pp, p¡0.001, 95% CI [+0.23, +0.81]. DocRED E-F1: +1.23pp, p¡0.001, 95% CI [+0.94, +1.55]. We report non-significant results transparently: SciERC F1 impact is neutral, not positive.

5 Related Work

Grammar-Constrained Decoding. XGrammar and Outlines compile context-free grammars for JSON schema enforcement. SynCode extends to pushdown automata. These handle structural (level-a) constraints but cannot express stateful type semantics. OntoCoDec's semantic layer addresses level-b (prefix-decidable with state) constraints, a fundamentally different decidability class.

Constrained Decoding for NLP. CRANE identifies quality degradation from over-constraining; our narrow constraint scope (only type tokens masked, free text unconstrained) avoids this. NeuroLogic and FUDGE apply soft/hard lexical constraints but lack ontology awareness.

LLM-Based IE. GoLLIE, CodeIE, and UniRel use code-style prompting for structured extraction. InstructUIE and DeepKE demonstrate instruction-tuned IE. None enforce type constraints at decoding time.

Ontology-Aware IE. OntoType and ontology-guided NER use ontology structure for entity typing but operate post-hoc or during training. OntoCoDec is the first to compile ontology constraints directly into the decoding process.

6 Conclusion

OntoCoDec demonstrates that ontology type constraints can be automatically compiled into efficient decoding constraints, achieving near-complete DR violation elimination (94% on SciERC, 44–90% on DocRED depending on model RV baseline) at zero or positive F1 impact. The framework generalizes across model families without code modification and is complementary to fine-tuning.

Limitations. Cross-field consistency violations (CF) increase from $\sim$0.1% to 9.9% (SciERC) under full constraint enforcement. This is an inherent limitation of per-relation independent type enforcement: the semantic layer assigns locally optimal types without global entity-level consensus. CF violations represent soft preference violations (individual triples remain type-valid) rather than hard structural violations, but they indicate room for improvement. Future work should explore cross-relation global consistency mechanisms (entity-level type consensus before relation-specific enforcement). Additionally, our evaluation covers 2 datasets and 3 models of similar scale (7–8B parameters); validation on larger models and diverse IE schemas would strengthen generalization claims.